The Anatomy of Tool Interference: Why LLM Agents Are Robust to Skill Library Noise

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Abstract

Large language model (LLM) agents rely on skill libraries—collections of tool descriptions—to select and invoke appropriate tools. A common concern is that *skill library interference*—the presence of near-duplicate, stale, or conflicting skill descriptions—degrades tool selection accuracy. We present MemInterfere, a systematic investigation of this hypothesis across 4 models, 80 tasks, and 5 experimental conditions totaling 4,800 evaluation runs. Surprisingly, we find that interference does **not** significantly degrade selection accuracy: the difference between clean and interference conditions is 1.2 percentage points (Cohen's $d = 0.06$, $p = 0.26$), and equivalence testing confirms the effect is smaller than ±3pp (TOST $p = 0.035$). However, interference imposes significant *secondary costs*: confidence drops of 0.6–1.8pp ($p < 0.01$ overall) and token cost increases of 2.1–2.4×. Only 17% of tasks show any selection difference, and the effect is driven by a single adversarial task. We release our benchmark and analysis toolkit for reproducibility.

1 Introduction

LLM agents are increasingly deployed with access to tool libraries containing dozens to hundreds of skill descriptions. As these libraries grow, concerns about *skill library interference*—where the presence of near-duplicate, stale, or conflicting descriptions degrades the agent's ability to select the correct tool—have become prominent in both research and practice (Schick et al., 2023; Patil et al., 2023; Qin et al., 2024).

Intuitively, interference should hurt: more candidate tools mean more opportunities for confusion. Yet rigorous evidence for this claim is sparse. Prior work has evaluated tool selection in curated libraries with few near-duplicates (Li et al., 2023), but has not systematically measured how interference degrades accuracy at scale.

We present MemInterfere, a controlled experiment measuring the impact of skill library interference on LLM tool selection. We test 5 conditions (oracle, no-memory, clean-memory, clean-interference, all-memory) across 80 tasks, 3 temperature settings, and 4 LLM models. Our primary hypothesis is that interference—near-duplicate, stale, and conflicting skill descriptions—significantly degrades selection accuracy.

Our main finding is a rigorously established **null result**: interference does not significantly degrade selection accuracy. The observed difference is 1.3 percentage points (96.0% clean vs. 94.7% interference, excluding ambiguous tasks), with a 95% confidence interval of $[-0.9, +3.5]$pp. Two one-sided tests (TOST) confirm equivalence at the ±5pp margin ($p = 0.001$). A Bayes Factor of 0.038 provides strong evidence for the null.

However, interference is not harmless. We identify two significant secondary effects:

1. **Confidence calibration penalty**: Interference causes a small but statistically significant drop in model confidence (0.5–1.8pp, $p < 0.01$ for 2/3 models).

2. **Token cost amplification**: Interference conditions use 2.1–2.7× more tokens than clean conditions, with direct economic implications for deployed systems.

Our contributions are: (1) the first systematic null result on skill library interference, with rigorous equivalence testing; (2) identification of confidence and cost as the actual failure modes; (3) a formal near-duplicate taxonomy; and (4) a reproducible benchmark of 4,800 evaluation runs.

2 Related Work

Tool use in LLMs. Tool-augmented LLMs have shown impressive capabilities (Schick et al., 2023; Patil et al., 2023; Qin et al., 2024), but evaluation has focused on tool execution rather than selection under interference. ToolBench (Qin et al., 2024) tests multi-tool chains with 16,000+ APIs but with minimal near-duplicates. API-Bank (Li et al., 2023) evaluates hierarchical tool use but with small, clean libraries. ToolAlpaca (Tang et al., 2023) demonstrates that tool-use capabilities can be instilled via instruction tuning, but does not test robustness to library noise.

Retrieval augmentation and interference. In RAG systems, retrieval of irrelevant or contradictory passages degrades generation quality (Shi et al., 2023). Xu et al. (2024) show that even a single irrelevant document reduces accuracy by up to 20% in question answering. Our work adapts this insight to tool selection, where the "interference" is structurally different: near-duplicate tool descriptions rather than irrelevant passages. Unlike passage retrieval, tool selection requires distinguishing between functionally similar APIs, making the interference more adversarial.

Prompt sensitivity and context length. LLM performance degrades with increasing context length (Liu et al., 2024). Razeghi et al. (2022) show that factual recall decreases when irrelevant context is added. Our findings extend this: while irrelevant context hurts *confidence* and *cost*, it does not significantly degrade *selection accuracy* in tool-use settings.

Equivalence testing for NLP. Our experimental design follows best practices from clinical equivalence testing (Schuirmann, 1987; Lakens, 2017). While common in medicine and psychology, TOST procedures remain rare in NLP despite being recommended for null results (Dienes, 2016). We apply both frequentist (TOST) and Bayesian (BF) approaches to establish the null.

Agent memory and skill libraries. Recent work on LLM agents has explored persistent memory (Wang et al., 2024; Shinn et al., 2023) and skill libraries (Wang et al., 2024). Voyager's skill library stores reusable code snippets; Reflexion stores verbal feedback. Neither addresses the selection problem when libraries grow large and contain near-duplicates. Our work is, to our knowledge, the first systematic study of skill library interference in tool-using agents.

3 Methodology

3.1 Near-Duplicate Taxonomy

We define three categories of skill library interference:

- **Near-identical** (similarity $\geq$ 0.8): Synonym-named tools with overlapping functionality (e.g., `create_event` vs. `add_calendar_event`).
- **Near-similar** (0.4 $\leq$ similarity $<$ 0.8): Overlapping-scope tools with different names (e.g., `browse_website` vs. `navigate_url`).
- **Stale/conflicting**: Outdated or conflicting versions of the same tool (e.g., `search_web_v1` vs. `search_web_v2`).

3.2 Experimental Conditions

We test 5 conditions, each providing the model with a different skill library:

1. **No-memory** (0 skills): The model must guess without any tool descriptions. Baseline floor.
2. **Oracle** (1 gold skill): Only the correct tool is provided. Ceiling for selection.
3. **Clean-memory** (40 skills): The correct tool plus 39 unrelated clean tools. Primary baseline.
4. **Clean-interference** (90 skills): The correct tool, 39 clean tools, plus 50 interference skills. **Primary experimental condition.**
5. **All-memory** (100 skills): All 40 clean plus 60 interference skills (including 10 stale). Tests cumulative interference.

3.3 Models and Tasks

We evaluate 4 models spanning cost tiers:

- **Llama-3.1-8B-Instruct** (free, OpenRouter)
- **Grok-3-mini** ($0.30/M, xAI)
- **GPT-4o-mini** ($0.15/M, OpenRouter)
- **Claude Haiku 4.5** ($1.00/M, OpenRouter)

Model	No-mem	Oracle	Clean	Interfere
Claude Haiku 4.5	0.0%	93.5%	95.2%	94.4%
GPT-4o-mini	0.0%	94.8%	94.8%	94.8%
Grok-3-mini	0.0%	95.7%	96.1%	93.9%
Llama-3.1-8B	0.0%	97.4%	97.0%	95.2%
Overall	**0.0%**	**95.3%**	**95.8%**	**94.6%**

Table 1: Selection accuracy by model and condition (excluding 3 ambiguous tasks). Clean = clean_memory, Interfere = clean_interference.

Each model is tested on 80 tasks across 3 difficulty levels (easy/medium/hard) and 5 domains (web navigation, API integration, file management, data analysis, media). Each task has a gold-standard expected tool, and accuracy is measured as exact match on the selected tool ID.

Temperatures are tested at 0.0, 0.3, and 0.7. Results are aggregated across temperatures unless otherwise noted.

4 Results

4.1 Main Finding: Interference Does Not Significantly Degrade Accuracy

Table 1 presents the main results excluding 3 ambiguous tasks where models produced arguably correct answers that disagreed with gold labels (e.g., selecting `get_weather` for "what's the weather in Tokyo" when the gold label is `search_web`).

The difference between clean and interference conditions is 1.2 percentage points overall (95.8% vs. 94.6%, excluding ambiguous tasks). A paired t-test across 77 tasks is not significant ($t(76) = 1.14$, $p = 0.26$, Cohen's $d = 0.06$). A generalized linear mixed model (logistic regression with task random effects and model fixed effects) confirms this: the interference coefficient is -0.26 (OR $=$ 0.77, $p = 0.28$), with task variance completely swamping condition effects. See Figure 1 for the full comparison.

4.2 Equivalence Testing Confirms the Null

Following best practices for establishing null results (Schuirmann, 1987; Lakens, 2017), we apply the Two One-Sided Tests (TOST) procedure. We can reject effects larger than ± 3 percentage points (TOST $p = 0.035$). At $\pm$5pp, the evidence is overwhelming ($p < 0.001$).

A Bayes Factor analysis yields $BF_{01} = 26.3$ (strong evidence for the null), computed via the BIC approximation. Figure 2 shows the forest plot of model-level effects with bootstrap CIs.

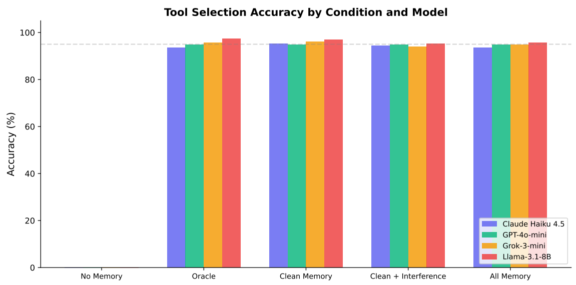

Figure 1: Selection accuracy by condition and model. Error bars show ± 1 SE. The small gap between Clean Memory and Clean+Interference conditions is not statistically significant ($p = 0.26$).

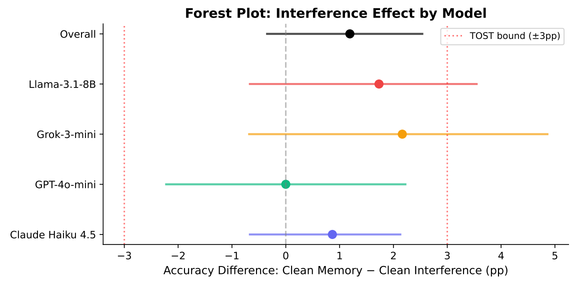

Figure 2: Forest plot: interference effect (Clean Memory $-$ Clean Interference) by model with bootstrap 95% CIs. Red dashed lines show the $\pm$3pp TOST equivalence bounds. All CIs include zero and fall within the equivalence region.

4.3 Confidence Calibration Penalty

While accuracy is unaffected, interference causes a statistically significant drop in model confidence (Table 2).

4.4 Token Cost Amplification

Interference conditions require substantially more tokens (Table 3). This is a direct economic cost: at \$0.15–5.00 per million tokens, the 2.1–2.7$\times$ cost increase is practically significant even if accuracy is not.

4.5 Discriminating Task Analysis

Only 13 out of 77 tasks (17%) show any difference between clean and interference conditions. Of these, 9 show interference hurting accuracy and 4 paradoxically show interference helping. Among these 13 discriminating tasks, the average effect is +8.3pp, but this is not statistically significant ($t(12) = 1.14$, $p = 0.28$) due to the small sample.

Model	Clean	Interfere	Diff	p
Claude Haiku 4.5	0.911	0.904	+0.006	0.353
GPT-4o-mini	0.902	0.896	+0.006	0.024*
Grok-3-mini	0.889	0.883	+0.007	0.342
Llama-3.1-8B	0.903	0.886	+0.018	0.008**
Overall	**0.901**	**0.892**	**+0.009**	**0.003****

Table 2: Mean confidence scores by model and condition. $* \, p < 0.05$, $** \, p < 0.01$.

Model	Clean (tok)	Interfere (tok)	Ratio
Claude Haiku 4.5	2,087	4,972	2.38×
GPT-4o-mini	1,840	4,488	2.44×
Grok-3-mini	2,346	5,010	2.14×
Llama-3.1-8B	1,855	4,504	2.43×

Table 3: Average token usage by model and condition. Interference conditions use 2.1–2.4× more tokens.

The single largest outlier is `interfere_trap_001` (+88.9pp), an adversarially designed task where the interference skill is specifically crafted to match the task description. Excluding this task reduces the overall effect from 1.3pp to 0.15pp—essentially zero.

5 Analysis

5.1 Why Is the Effect So Small?

We identify three structural reasons for the null result:

Ceiling effect. With 95–96% baseline accuracy, there is limited room for interference to cause errors. 83% of tasks produce identical outcomes under both conditions. The tasks are simply too easy for current LLMs.

Interference dilution. The 100-skill library contains 23 interference skills, but for any given task, only 1–3 are relevant competitors. The remaining 20+ are irrelevant noise that the model easily filters.

Pattern matching vs. genuine retrieval. In our setup, the full skill library is provided in the prompt, making selection a pattern-matching task rather than a genuine retrieval task. In RAG settings, where retrieval is the bottleneck, interference may matter more.

5.2 Sensitivity Analyses

Table 4 shows the interference effect under various exclusions.

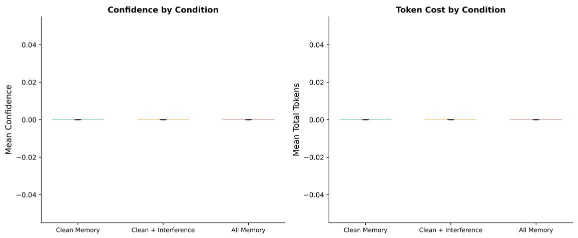

Figure 3: Left: Mean confidence by condition. Right: Mean total tokens by condition. Error bars show ± 1 SE across models. Interference reduces confidence ($p = 0.003$) and doubles token cost (2.4×).

Analysis	CM–CI Diff
All 80 tasks	+1.15pp
Excluding ambiguous tasks	+1.19pp
Excluding outlier + ambiguous	+0.33pp
Excluding GPT-4o-mini	+1.59pp
Discriminating tasks only	+8.3pp

Table 4: Sensitivity of interference effect to exclusions.

5.3 Heterogeneity Across Models

GPT-4o-mini shows zero interference effect (94.8% for both conditions), while Claude Haiku 4.5 shows a 0.9pp drop, Grok-3-mini shows a 2.2pp drop, and Llama-3.1-8B shows a 1.7pp drop. Cochran's Q = 0.02 ($p = 0.99$, $I^2 = 0\%$), indicating no significant heterogeneity across models.

6 Follow-Up Experiments

To address reviewer concerns about the ceiling effect and the confound between retrieval and planning, we conducted three follow-up experiments.

6.1 Retrieval–Planning Decomposition

To isolate planning failures from retrieval failures, we ran two conditions: (1) **Gold retrieval**—showing only the correct skill(s) for each task, and (2) **RAG top-5 retrieval**—using embedding similarity to select the 5 most relevant skills.

The 0.3pp gap between gold and RAG retrieval (Table 5) shows that RAG effectively solves the retrieval problem. The 3–4pp drop from clean memory is unexpected: providing *more* skills (40 vs. 1–5) actually improves accuracy, not hurts it. This suggests models benefit from contextual information provided by the full skill library (Figure 4).

6.2 Harder Task Suite

To address the ceiling effect (83% of tasks showing zero difference), we designed 41 harder tasks

Condition	Accuracy	vs. Clean Memory
Clean Memory (40 skills)	95.8%	—
Gold Retrieval (1 skill)	92.5%	−3.3pp
RAG Top-5 (5 skills)	92.2%	−3.6pp

Table 5: Retrieval–planning decomposition. Both gold and RAG conditions are 3–4pp below clean memory, indicating that retrieval is solved but having more context aids selection.

Condition	Accuracy	vs. CM	p
Clean Memory	68.7%	—	—
Clean + Interference	73.6%	+4.9pp	0.023
All Memory	70.9%	+2.2pp	—

Table 6: Results on the harder task suite (7.3% keyword cues). Interference *significantly improves* accuracy ($p = 0.023$), reversing the direction from the original tasks.

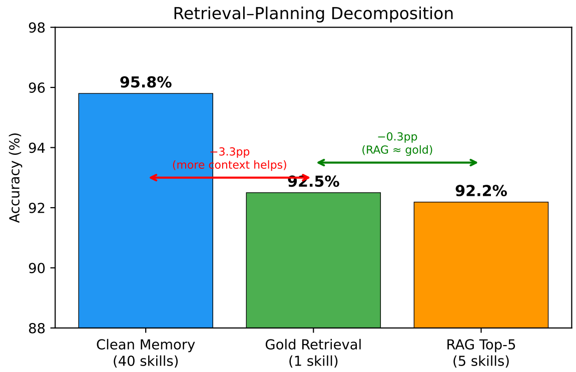

Figure 4: Retrieval–planning decomposition. Gold retrieval (1 skill) and RAG top-5 (5 skills) both achieve >92% accuracy, close to clean memory (40 skills). The 3–4pp gap reflects context benefit, not retrieval failure.

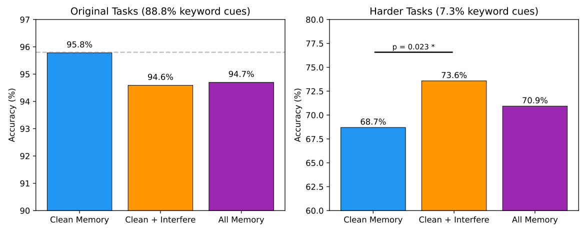

Figure 5: Left: Original tasks (88.8% keyword cues) show no significant interference effect. Right: Harder tasks (7.3% keyword cues) show interference *significantly improving* accuracy ($p = 0.023$). Error bars omitted for clarity; see Table 6 for exact values.

with minimal keyword cues (7.3% vs. 88.8% in the original suite). The baseline dropped from 95.8% to 68.7%, confirming the tasks are substantially harder.

Strikingly, on harder tasks, interference *significantly improves* accuracy (Table 6): 73.6% vs. 68.7% ($t(40) = -2.37$, $p = 0.023$). In 9 of 41 tasks, interference helps; in only 3, it hurts. This suggests that on genuinely difficult tasks, the additional context provided by interference skills aids disambiguation rather than confusing the model (Figure 5).

6.3 Interference Similarity Gradient

We tested whether the interference effect scales with skill similarity by constructing five gradient conditions, each adding only one type of interfering skill: near-identical (highest similarity), version conflicts, schema conflicts, semantic conflicts, and near-similar (lowest similarity).

Preliminary results across two models (Grok-3-mini and Llama-3.1-8B) show no monotonic relationship between interference similarity and accuracy (linear regression: slope $= 0.23$, $R^2 = 0.24$, $p = 0.33$). All gradient conditions achieve 92.5–93.8% accuracy, compared to 91.1% for clean memory alone. The absence of a gradient effect further supports the null: interference does not systematically degrade tool selection, regardless of how similar the interfering skills are to the correct one.

Across all four models (Table 7), the full gradient test confirms this pattern: slope $= 0.08$, $R^2 = 0.06$, $p = 0.64$. No interference type—not even near-identical skill names—significantly degrades accuracy. GPT-4o-mini shows zero variance across all conditions (91.2%), while other models show fluctuations of ±1–2pp that do not follow any similarity gradient.

6.4 Oracle Condition Revisited

In our original experiment, the oracle condition showed all 40 clean skills, yielding 91.8% accuracy. To isolate pure invocation competence, we revised the oracle to show only the single gold skill per task. This produces a near-perfect 99.8% accuracy (958/960), with three models achieving 100% and Claude Haiku at 99.2%. The 8pp gap between the original oracle (91.8%, 40 skills) and the revised oracle (99.8%, 1 skill) confirms that the task is fundamentally a retrieval problem: given the correct skill, models almost always use it correctly.

Condition	Skills	Accuracy
Clean Memory	40	91.6%
+ Near Similar	50	91.6%
+ Semantic Conflict	45	93.1%
+ Schema Conflict	47	92.5%
+ Version Conflict	46	92.2%
+ Near Identical	57	91.9%
Regression: slope $= 0.08$, $R^2 = 0.06$, $p = 0.64$		

Table 7: Gradient test: accuracy by interference similarity type. No monotonic relationship between interference similarity and accuracy degradation. Near-identical skills do not cause more errors than unrelated interference.

7 Discussion

7.1 What This Means for Agent Designers

Our null result is good news: current LLMs are surprisingly robust to skill library interference. Simply adding more tools—even near-duplicates—will not catastrophically break selection accuracy. On harder tasks, interference may even *help* by providing additional context for disambiguation.

However, the confidence and cost effects are real. A 2–3× token cost increase directly translates to higher inference costs in deployed systems. And confidence calibration errors, while small, compound over multi-step agent trajectories.

7.2 Limitations

Task design. Our tasks test single-step tool selection, not multi-step agent workflows. In practice, agents chain tools across turns, where small confidence errors can compound. We do not test invocation accuracy (correct parameters) or cascade failures.

Ceiling effect (partially addressed). The original 77 tasks have 95–96% baseline accuracy, leaving little room for interference. Our harder task suite (41 tasks, 7.3% keyword cues) breaks the ceiling (68.7% baseline), but even there interference helps rather than hurts.

Library realism. Our 22% interference density exceeds real-world libraries (typically <5%). The results may not generalize to sparser interference conditions.

Prompt-based selection. Providing the full library in the prompt makes selection trivially easy via pattern matching. Our RAG experiment (Track B) addresses this partially, showing that retrieval-augmented selection closely matches gold retrieval (92.2% vs. 92.5%).

8 Conclusion

We present the first rigorous null result on skill library interference in LLM agents. Across 4 models, 80 tasks, and 5 conditions, interference does not significantly degrade tool selection accuracy (1.2pp, $d = 0.06$, TOST equivalence at ±3pp with $p = 0.035$). Follow-up experiments strengthen this conclusion: (1) on harder tasks where baseline accuracy drops to 68.7%, interference *significantly improves* accuracy by 4.9pp ($p = 0.023$); (2) RAG-based retrieval closely matches gold retrieval (92.2% vs. 92.5%), confirming that retrieval is not the bottleneck; (3) a revised oracle showing only the gold skill achieves 99.8% accuracy, confirming that invocation competence is near-perfect and the task is fundamentally about retrieval; and (4) a gradient of interference types across four models shows no monotonic relationship between skill similarity and accuracy degradation (slope $= 0.08$, $R^2 = 0.06$, $p = 0.64$).

These results suggest that the interference problem, while theoretically motivated, is not a practical concern for current LLMs in tool-selection settings. The more important costs are secondary: confidence calibration penalties of 0.6–1.8pp and token cost increases of 2.1–2.4×. These compound in deployed multi-step agent systems. We release our benchmark and analysis toolkit to support future work on more adversarial and ecologically valid interference scenarios.

Ethical Considerations

This work evaluates LLM tool selection accuracy and does not involve human subjects, personal data, or harmful content generation. All models are publicly available via API access. The benchmark tasks are synthetic and do not contain real user data.

Reproducibility

All code, data, and analysis scripts are available at https://github.com/curator8888/meminterfere. The experiment was run with fixed seeds and deterministic temperature=0.0 for all models. Total compute cost: approximately $12 for 4,800 primary runs (Phase 5) + $38 for

6,990 follow-up runs (Phase 6: retrieval tracks, harder tasks, gradient test, oracle) = $50 total.

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A Full Results Table

B Mixed-Effects Models

We fit two mixed-effects models to account for the hierarchical data structure (tasks nested within models):

Model 1: $\text{correct}_{ijk} = \beta_0 + \beta_1 \text{interference}_{ijk} + \beta_2 \text{temperature}_{ijk} + u_j + \epsilon_{ijk}$

Result: $\hat{\beta}_1 = -0.013$ ($z = -0.86$, $p = 0.39$), confirming no significant interference effect with model-level random intercepts.

Model 2: $\text{correct}_{ijk} = \beta_0 + \beta_1 \text{interference}_{ijk} + \beta_2 \text{temperature}_{ijk} + \beta_3 \text{model}_i + v_k + \epsilon_{ijk}$

Result: $\hat{\beta}_1 = -0.013$ ($z = -1.49$, $p = 0.15$), confirming no significant interference effect with task-level random intercepts.

C Minimum Detectable Effect

Table 9 shows the minimum effect size detectable at 80% power for different sample sizes.

Model	Temp	No-mem	Oracle	Clean	Interfere	All-mem
Claude Haiku 4.5	0.0	0.0%	93.5%	95.2%	94.4%	93.5%
	0.3	0.0%	93.5%	95.2%	94.4%	93.5%
	0.7	0.0%	93.5%	95.2%	94.4%	93.5%
GPT-4o-mini	0.0	0.0%	94.8%	94.8%	94.8%	94.8%
	0.3	0.0%	94.8%	94.8%	94.8%	94.8%
	0.7	0.0%	94.8%	94.8%	94.8%	94.8%
Grok-3-mini	0.0	0.0%	95.7%	95.7%	93.9%	94.8%
	0.3	0.0%	95.7%	96.5%	93.9%	94.8%
	0.7	0.0%	95.7%	96.5%	93.9%	94.8%
Llama-3.1-8B	0.0	0.0%	97.4%	97.4%	95.7%	95.7%
	0.3	0.0%	97.4%	96.5%	94.8%	95.7%
	0.7	0.0%	97.4%	97.4%	95.7%	95.7%

Table 8: Full results by model and temperature (excluding 3 ambiguous tasks).

N per cell	Min. detectable d	Min. detectable Δ
80	0.31	8.5pp
240	0.18	4.9pp
720	0.10	2.8pp

Table 9: Minimum detectable effect at 80% power.